

IBM BOB HACKATHON • TRACK: AI • TURN IDEA INTO IMPACT FASTER

CrisisAI

Emergency Incident Prioritization & Deduplication Dashboard

Turning fragmented, unstructured emergency reports into a single, prioritized operational picture — so dispatchers know which incident needs attention first, and why.

TEAM TECHTITANS • TRACK: AI

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THE PROBLEM

Chaos in the first minutes of a crisis



WHAT

Fragmented, duplicate reports

During disasters, dispatchers are flooded with unstructured reports from citizens, SMS, hotlines, and field teams — many describing the exact same incident.



WHO

Dispatchers & incident commanders

The people who must manually read, categorize, and cross-reference every single report before deciding who to send, and where.



WHY IT MATTERS

Minutes cost lives

Keyword-based, manual triage misses the difference between a routine observation and trapped, vulnerable people — and duplicate tickets waste scarce response capacity.

11

separate citizen reports were filed for one flooded railway station in our demo — before deduplication. Without it, dispatchers see 11 tickets instead of 1 real emergency.

OUR SOLUTION

One command center, built from thousands of signals

CrisisAI converts unstructured citizen reports into structured, prioritized, deduplicated incidents — and presents them through a live Emergency Command Center.

1

Ingest

Multi-channel reports (SMS, web, mobile, radio)

2

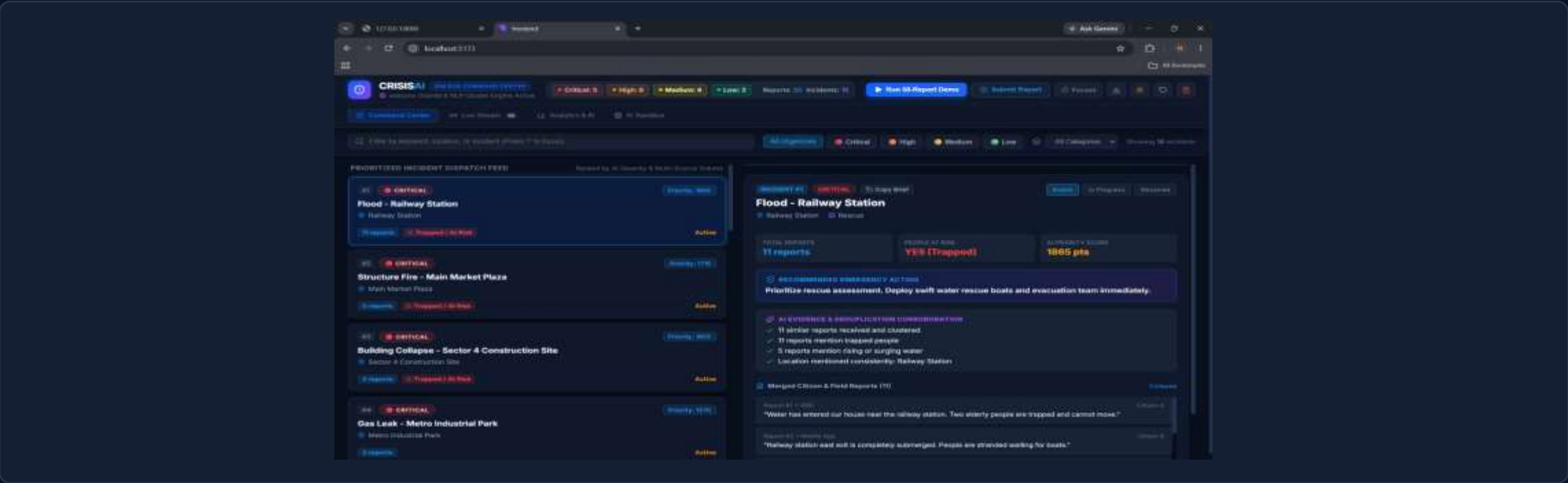
Deduplicate

Cluster corroborating reports automatically

3

Prioritize & Act

Rank by human-life risk, dispatch with confidence



How CrisisAI works under the hood



MULTI-FACTOR PRIORITY SCORE

Priority = **Urgency** + Category Weight + Human Risk Bonus
+ $\log_2(\text{Reports} + 1) \times 60$

Rewards corroboration by multiple sources, without letting report volume alone outrank an immediate life-threatening event.

DEDUPLICATION DECISION

IF **cosine similarity** ≥ 0.55
AND location / type are corroborated

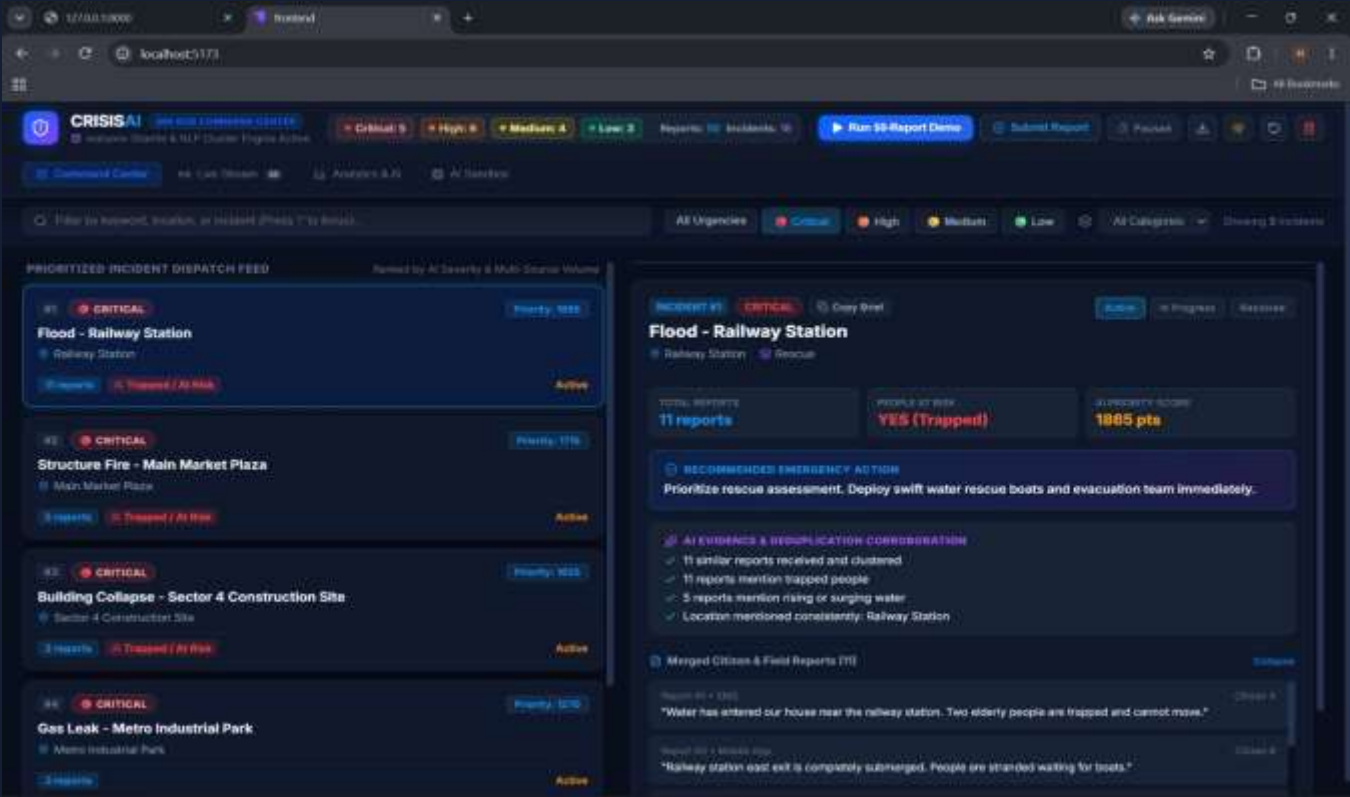
✓ **Merge report into existing incident**

✗ Otherwise → create a new incident

Threshold configurable via SIMILARITY_THRESHOLD.

KEY FEATURE

Watch 11 reports become 1 ranked incident



The screenshot displays the CRISIS AI interface. On the left, a 'PRIORITIZED INCIDENT DISPATCH FEED' lists four incidents: 'Flood - Railway Station' (Priority: 1865, Critical), 'Structure Fire - Main Market Plaza' (Priority: 1716, Critical), 'Building Collapse - Sector 4 Construction Site' (Priority: 1624, Critical), and 'Gas Leak - Metro Industrial Park' (Priority: 1570, Critical). On the right, a detailed view of 'INCIDENT #1: CRITICAL' for 'Flood - Railway Station' is shown. It indicates '11 reports' and '1865 pts' priority score. A 'RECOMMENDED EMERGENCY ACTION' box suggests prioritizing rescue assessment and deploying water rescue boats. A 'Merged Citizen & Field Reports (11)' section shows two reports: one from a citizen stating 'Water has entered our house near the railway station. Two elderly people are trapped and cannot move.' and another from a witness stating 'Railway station east exit is completely submerged. People are stranded waiting for boats.'

Incident #1 — Flood, Railway Station (Priority 1865, Critical)

- 1

11 reports arrive

Citizens report flooding near the railway station across SMS, web, and the mobile app.
- 2

AI clusters them

TF-IDF + cosine similarity merges all 11 into a single incident, not 11 tickets.
- 3

Risk is flagged

“Trapped” and “elderly” language is detected — People at Risk: Yes.
- 4

Action is generated

AI recommends: deploy water-rescue and evacuation teams immediately.

Powered by IBM watsonx.ai Granite



Granite 3.0 for NLP Extraction

ibm/granite-3-8b-instruct performs structured entity extraction — incident type, location, urgency, people at risk — straight from raw citizen text.



watsonx.ai API Integration

Configured via WATSONX_APIKEY, WATSONX_PROJECT_ID and WATSONX_URL, calling the watsonx.ai inference endpoint for every incoming report.



Dual-Engine Resilience

AI_MODE=watsonx routes extraction to Granite; if cloud credentials or connectivity aren't available, a local NLP engine keeps the system fully operational.



Live in the Dashboard

The Analytics view shows Watsonx Granite NLP status and latency in real time, so operators can see the model working end-to-end.



“Watsonx Granite NLP: Active (24ms Latency)” — visible live in the Analytics & AI tab, alongside Watsonx-ranked severity and category breakdowns.

RESULTS & IMPACT

50 raw reports → 18 actionable incidents

64%

Clutter Cut

5

Critical Incidents

6

High Priority

11

Reports Merged
into Top Incident

HIGHLIGHTED INCIDENT

Railway Station Flood — 11 reports merged into 1

- ✓ Urgency: Critical · Vulnerable elderly residents reported trapped
- ✓ Corroborating evidence: multiple reports cite trapped people & rising water
- ✓ AI-recommended action: prioritize rescue assessment, deploy water-rescue and evacuation teams

WHAT WE'RE MOST PROUD OF

A complete end-to-end intelligence pipeline — not isolated tickets, but one operational picture, from noisy messages to ranked, dispatch-ready incidents.

“Which incident needs attention first, and why?”

TEAM

Team TechTitans



Janvi Prajapati

Team Lead & AI Engineer

Led the Watsonx Granite integration and built the multi-factor priority scoring engine.



Krima Parmar

Backend Engineer

Built the FastAPI services and the TF-IDF + cosine similarity deduplication engine.



Hemangi Parmar

Frontend Engineer

Built the React command center — live stream, incident feed, and tactical geo-grid.



Vidhi Patel

AI / QA Engineer

Built the NLP extraction pipeline's local fallback and the automated test & evaluation suite.